AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

R26-SE-025

Project Proposal Report

Senarathna A.M – IT22262554

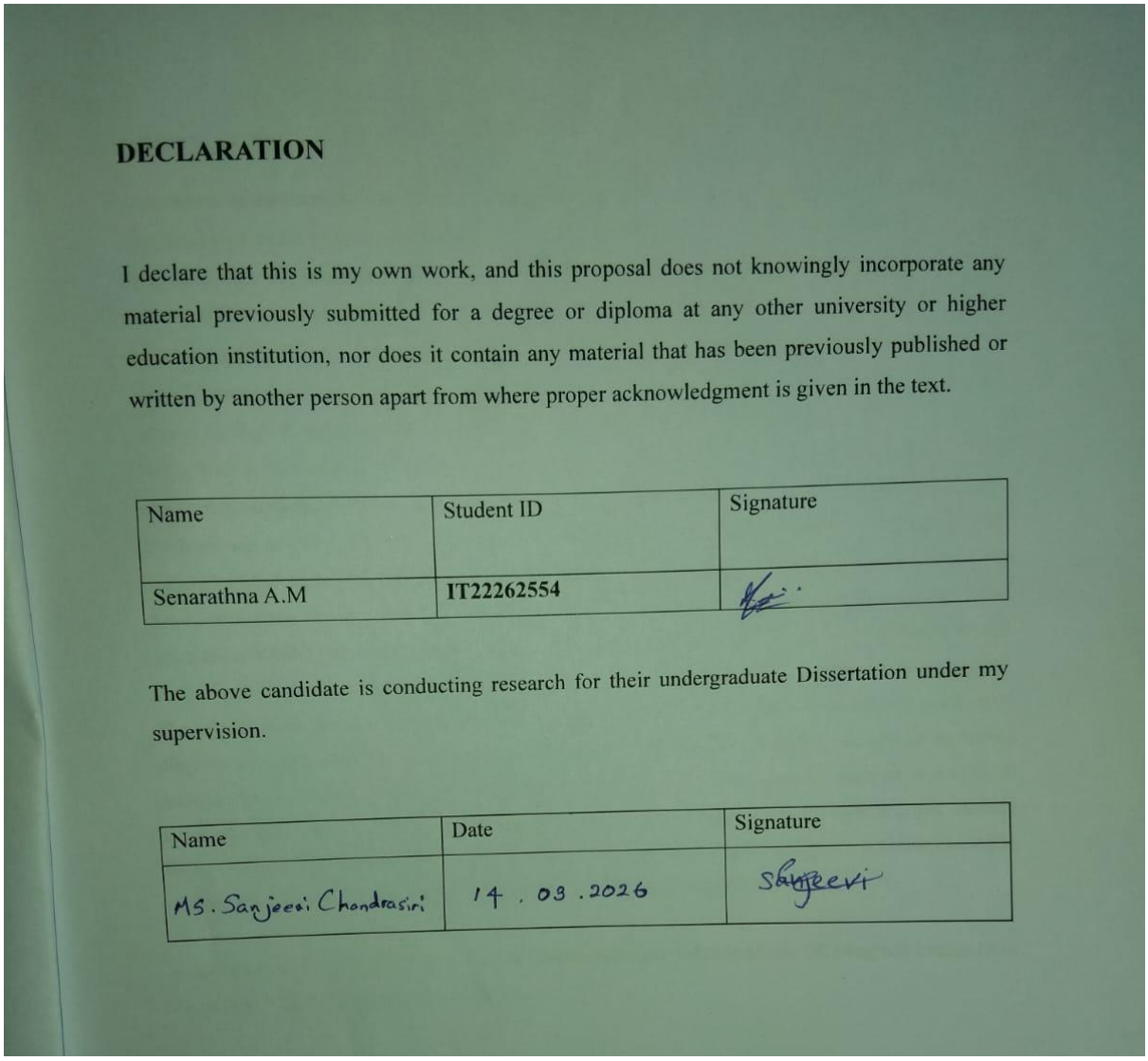
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Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION



DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Senarathna A.M	**IT22262554**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
MS. Sanjeevi Chandrasiri	14.03.2026	

ABSTRACT

Automated assessment in education is increasingly important as the number of students and the complexity of exam answers continue to grow. Traditional manual grading methods may introduce inconsistencies due to subjective factors, grading fatigue, and differences in teacher judgment, which can affect fairness and reliability in academic evaluation. These challenges become more significant when assessing diagram-based answers such as Entity–Relationship (ER) diagrams, which require careful interpretation of structures and relationships. Existing research mainly focuses on diagrams created in digital environments or relies on stroke reconstruction techniques that may introduce recognition errors, and most systems do not support diagrams drawn on real exam answer sheets containing background elements such as rulers or paper lines. To address this limitation, this research proposes an AI-Integrated Multi-Modal Academic Evaluation system capable of extracting and evaluating handwritten ER diagrams directly from scanned or photographed student answer sheets. The proposed system applies image preprocessing to remove background noise and paper lines, object detection models such as YOLO and Faster R-CNN to locate diagram regions, and convolutional neural networks to recognize diagram components including entities, relationships, attributes, and connectors. The extracted diagram is then transformed into a semantic representation and compared with a reference diagram from the marking scheme using semantic equivalence rules, allowing the system to accept multiple valid diagram solutions and perform dynamic grading. The expected outcome is a reliable automated assessment tool that improves grading consistency, reduces lecturer workload, and enables faster evaluation of diagram-based exam answers. The system also aims to support scalable academic assessment processes while maintaining fairness and accuracy in grading.

Keywords: Automated Assessment, Handwritten Diagram Recognition, ER Diagram Evaluation, Academic Grading Automation, AI in Education.

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LIST OF ABBREVIATIONS

Table 1 Abbreviation

Abbreviation	Full Form
AI	Artificial Intelligence
ER	Entity–Relationship
UML	Unified Modeling Language
HDE	Handwritten Diagram Extraction
CNN	Convolutional Neural Network
OCR	Optical Character Recognition
YOLO	You Only Look Once
R-CNN	Region-based Convolutional Neural Network
LLM	Large Language Model
GUI	Graphical User Interface
JSON	JavaScript Object Notation
WBS	Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

Assessing student achievement through teacher-based grading raises concerns about reliability and fairness due to the influence of non-academic and subjective factors. Accurate assessment is essential in education because it directly affects students' academic progress, opportunities, and overall evaluation of their learning outcomes. However, when grading methods are inconsistent or influenced by external factors, it can lead to unequal evaluation of students' work. Ensuring fairness and consistency in assessment is therefore important to maintain trust in educational systems and to provide students with an accurate reflection of their academic abilities. Evidence from research highlights how non-academic factors can influence grading decisions. For instance, teachers may sometimes award higher grades to students who demonstrate better behavior even when their academic performance is similar. Research also indicates that students may receive different grades based on gender despite having identical test scores. Additionally, grading consistency can vary significantly among teachers. [1] In a study analysis of teachers using a 100-point grading scale showed a wide variation in marks, with scores ranging from 50 to 96 for the same work. This variation highlights inconsistencies in teacher assessment. Other factors such as the halo effect also influence grading, where teachers may judge the entire answer based on a single impression, such as a strong introduction or poor handwriting. [2]

Workload and repetitive marking further affect grading reliability. In study demonstrates that boredom increases during long grading sessions, which can lead to decreasing marks over time. Their findings indicate that each increase in boredom reduces marks by approximately 2.8 points, meaning that later exam papers may receive lower scores compared to earlier ones. [3] At the same time, the growing demand for software engineers has led to increased student enrollment in computer science programs, creating additional pressure on educators to evaluate assignments efficiently. Assessing complex representations such as Unified Modeling Language diagrams requires considerable time and effort, making manual grading both expensive and time-consuming. These challenges highlight the need for automated assessment approaches. [4] However, evaluating open-ended design problems such as Entity Relationship (ER) diagrams and relational schemas remains a significant challenge due to the variability in valid representations. Manual evaluation in such contexts is neither scalable nor efficient. Hence, automating the assessment process becomes a necessity [5]

The proposed solution aligns with the Quality Education, which aims to ensure inclusive, equitable, and high-quality education. By developing a system to automatically assess handwritten exam ER diagrams, the solution helps improve fairness and consistency in grading by reducing the influence of subjective factors and grading fatigue. Automating the evaluation process can also reduce the workload on lecturers and enable faster feedback to students, supporting more effective learning and maintaining reliable academic assessment standards.

1.2 Literature Review

Existing Research in handwritten diagram recognition focuses on converting scanned or photographed diagrams into structured digital models by recognizing symbols and identifying their relationships. Approaches in this field are mainly categorized into online and offline recognition. Online methods capture temporal stroke data using devices such as tablets, while offline recognition works directly with raster images of hand-drawn diagrams on paper. Due to the limited availability of offline datasets, many studies adapt online datasets by rendering stroke data as images. [6] Existing research mainly evaluates diagrams drawn on clean digital canvases or controlled datasets and therefore does not fully address practical scenarios such as diagrams drawn on paper with background noise or answer sheet patterns.

In 2015 proposes a framework for offline sketch parsing that aims to recognize shapes and text from hand-drawn diagrams captured as images. The research introduces a sharpness estimation approach to efficiently identifying stroke groups that represent valid shapes before performing recognition, reducing the number of candidate groups and improving efficiency. The framework uses a three-stage cascade process consisting of sharpness estimation, shape recognition, and sketch parsing using domain knowledge. [7] This Evaluation does not account for stroke reconstruction errors, therefore the results cannot be directly compared with image based recognition systems. Some reconstruction can sometimes be incorrect. As a result, the performance cannot be fairly compared with systems that work directly on images.

Research in handwriting recognition follows two main approaches: on-line and off-line. Online recognition captures dynamic stroke data from devices like tablets, allowing higher precision, while off-line recognition works with static images, useful for historical documents or digitizing notes. In this research, strokes reconstruction method Each pixel of the skeleton is labeled as an ambiguous or unambiguous based on its 3×3 neighborhood [8] It cannot fully reconstruct original strokes, leading to potential decreases in recognition accuracy due to incorrect stroke structures. Intersections or overlapping strokes may result in incorrect

connections, and the method is intentionally simple, lacking advanced optimization. Performance also degrades in the presence of noise, as real-world inputs are less clean than datasets.

In 2017 propose a system with Enhancing Feedback to Students in Automated Diagram Assessment [9] but this research not handling the handwritten diagrams. Students must draw diagrams digitally inside the system does not support Handwritten diagrams, Paper exam diagrams, Scanned answer scripts, Camera-captured diagrams So it cannot be used for real exam papers.

Raji Ramachandran and her colleagues work on studying that AI Assisted System for Automated Evaluation of Entity-Relationship Diagram and Schema Diagram Using Large Language Models in 2025. [5]This research mainly focused on the generated ER within the system solution by given the exam case study and evaluating the students diagram with considering the ER Diagram generated by the system. This system operates using PlantUML, a text-based language for creating UML diagrams and Student Submissions: Students use a Graphical User Interface (GUI) to visually design their Entity-Relationship (ER) and schema diagrams, which are then automatically converted into PlantUML code for evaluation.

Recent research on automated UML diagram grading has explored multiple approaches to address the challenges of evaluating diverse and creative student solutions. Rule-based systems and heuristic methods provide consistent and fast grading but struggle with flexible or novel designs. Machine learning techniques improve adaptability and handle large datasets but face data dependency and high computational costs. [4] (Students did not submit handwritten UML diagrams Instead they used a digital UML tool that exports diagrams as PNG and JSON files. This research uses a non-semantic method. The pipeline converts the UML Diagram into the JSON format and then into Text Description and Compare with case study and grading

1.3 Research Gap

Existing research on automated diagram recognition and evaluation shows several limitations when applied to real handwritten exam environments. Many studies assume clean digital inputs rather than real paper-based student answers. Many Research Works approach relies on datasets created from clear drawing canvases and does not consider the background conditions of student answer sheets. [6] [7] [8] In real examinations, diagrams are drawn on ruled papers where horizontal lines and margins exist. These background elements must be separated from

the diagram before recognition. Additionally, diagram components such as arrows or connector lines may overlap with the paper rulers, making it difficult to distinguish between diagram structures and page markings. Current approaches do not address this preprocessing challenge.

Several offline sketch recognition studies attempt to reconstruct drawing strokes from static images instead of directly analyzing the diagram structure, but this reconstruction process can introduce errors. During evaluation, the system does not account for these stroke reconstruction errors, meaning the results cannot be fairly compared with systems that operate directly on image-based diagram. [7] Some works shows that reconstructed strokes may be fragmented into multiple small segments, especially when strokes intersect or overlap. [8] The algorithm is also relatively simple and performs poorly when noise is present.

Other automated diagram assessment systems mainly focus on diagrams created digitally within software tools rather than handwritten diagrams. The system proposed in research work evaluates diagrams generated through a graphical interface and converts them into PlantUML code. [5] [9]This approach assumes diagrams are digitally structured and may produce overly complex reference solutions compared to typical student answers. Additionally, the system initially relies on a single reference diagram, which limits its ability to evaluate alternative valid solutions. Additionally, the system initially relies on a single reference diagram, which limits its ability to evaluate alternative valid solutions.

Recent research explores automated grading using LLM-based methods, but it still relies on diagrams created using digital UML tools. The evaluation pipeline converts diagram structures into JSON and textual descriptions before grading, meaning it does not perform image-based diagram detection or handwriting recognition. And this study uses a relatively small dataset of 92 diagrams. Furthermore, the approach encourages students to produce similar UML designs, even though multiple valid diagram structures may exist for the same scenario.

Therefore, there is a clear need to develop methods that support the recognition and evaluation of handwritten diagrams drawn in real exam answer sheets. In the context of improving assessment practices and supporting Quality Education, there is space for research that can automatically extract and interpret diagrams from paper-based student submissions. Current approaches mainly rely on clean digital diagrams, structured diagram data, or imperfect stroke reconstruction techniques, which limit their applicability in real examination environments.

Table 2 Research Gap

Feature / Capability	R1	R2	R3	R4	R5	R6	Proposed System
Supports handwritten diagrams	Yes	Yes	Yes	No	No	No	Yes
Handles answer sheet background (rulers / paper lines)	No	No	No	No	No	No	Yes
Uses stroke reconstruction from images	No	Yes	Yes	No	No	No	No
Uses object detection–based diagram recognition	Yes	No	No	No	No	No	Yes
Works with scanned exam papers / real paper submissions	No	No	No	No	No	No	Yes
Uses semantic diagram understanding for grading	No	No	No	Yes	Yes	Yes	Yes
Supports multiple valid diagram solutions	No	No	No	No	No	No	Yes

2. OBJECTIVE

2.1 Main Objective

To develop a reliable and efficient automated system capable of extracting handwritten Entity–Relationship (ER) diagrams from scanned student answer sheets and accurately interpreting their structural components. The system will then evaluate the extracted diagram by comparing it with a predefined marking-scheme reference diagram to assess correctness and completeness. The proposed system aims to achieve high recognition accuracy while handling real exam conditions. Ultimately, this system is intended to support educators and academic institutions.

2.2 Specific Objectives

Handwritten Diagram Detection – Detect and segment ER diagrams from scanned or camera-captured student answer sheets, accounting for background elements such as rulers, lines, and other page artifacts.

Component Recognition – Identify and classify diagram components including entities, relationships, attributes, and arrows with high accuracy using image processing and OCR-based semantic detection.

Semantic Diagram Comparison and Evaluation – Compare the semantic representation of the student's diagram with the marking guide's reference diagram to assess correctness, completeness, structural accuracy, and adherence to ER modeling conventions.

Dynamic Scoring – Allow flexible grading for alternative but valid diagram designs, reflecting real-world variability in student solutions.

Error Handling and Noise Robustness – Handle incomplete, overlapping, or misdrawn strokes and reduce the impact of noise from scanned or photographed answer sheets.

3. METHODOLOGY

3.1 Research Approach

The study focuses on designing and developing an AI-based Handwritten ER Diagram Extraction (HDE) and automated assessment system for UML diagrams drawn in exam answer sheets. The proposed system consists of multiple processing stages. First, the answer sheet preprocessing stage handles the background structure of exam papers, particularly the printed rulers present on many answer sheets. The system measures the distance between rulers and removes or ignores these lines to generate a clean canvas for diagram processing. This step must also consider situations where students draw diagram lines along the ruler lines, as those lines could represent arrow connections or component boundaries. After preprocessing, the system performs handwritten diagram extraction from the scanned image. The system in this research focuses on image-based detection of diagram structures directly from the scanned answer sheets.

After identifying the diagram structure, the system performs automated diagram assessment by comparing the extracted diagram with a reference diagram derived from the marking scheme. The goal is to support dynamic grading, allowing valid alternative diagram designs rather than forcing students to match a single predefined diagram structure.

3.2 Tools and platforms with justification

Image Preprocessing Techniques - Image preprocessing is used to improve the quality of scanned exam papers by removing noise, paper lines, and shadows. This helps the system focus only on the handwritten content, making later recognition steps more accurate.

YOLO (You Only Look Once) - YOLO is used for fast object detection to identify and locate diagram regions in exam answer sheets. It processes the entire image at once, allowing quick and efficient detection of diagrams.

Faster R-CNN - Faster R-CNN provides highly accurate object detection by generating regional proposals and classifying them. It is useful when precise localization of diagram areas is required in complex exam papers.

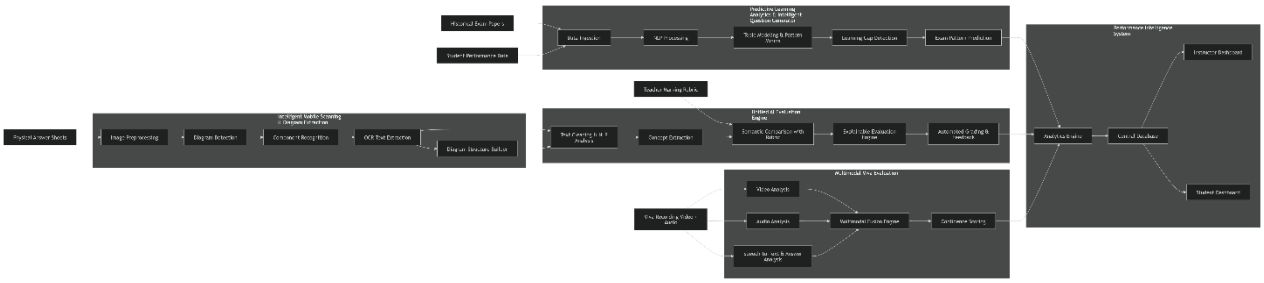
Convolutional Neural Networks (CNNs) - CNNs are used to recognize handwritten diagram symbols and text labels. They learn visual patterns from training data, allowing the system to classify different diagram elements accurately.

Semantic Equivalence Rules - Semantic equivalence rules allow the system to accept different diagram structures that represent the same meaning. This ensures fair assessment even when students draw diagrams in slightly different but logically correct ways.

3.3 Data collection and ethical considerations

The dataset for this research will consist of handwritten Entity–Relationship (ER) diagrams drawn by students in past examination papers. These diagrams will be collected from previously conducted exam answer sheets where students were required to design ER diagrams. The diagram sections will be captured as images using scanning or camera-based digitization, and only the diagram regions will be extracted. This data will be used for training and testing the handwritten diagram extraction and assessment system.

4. HIGH LEVEL SYSTEM ARCHITECTURE



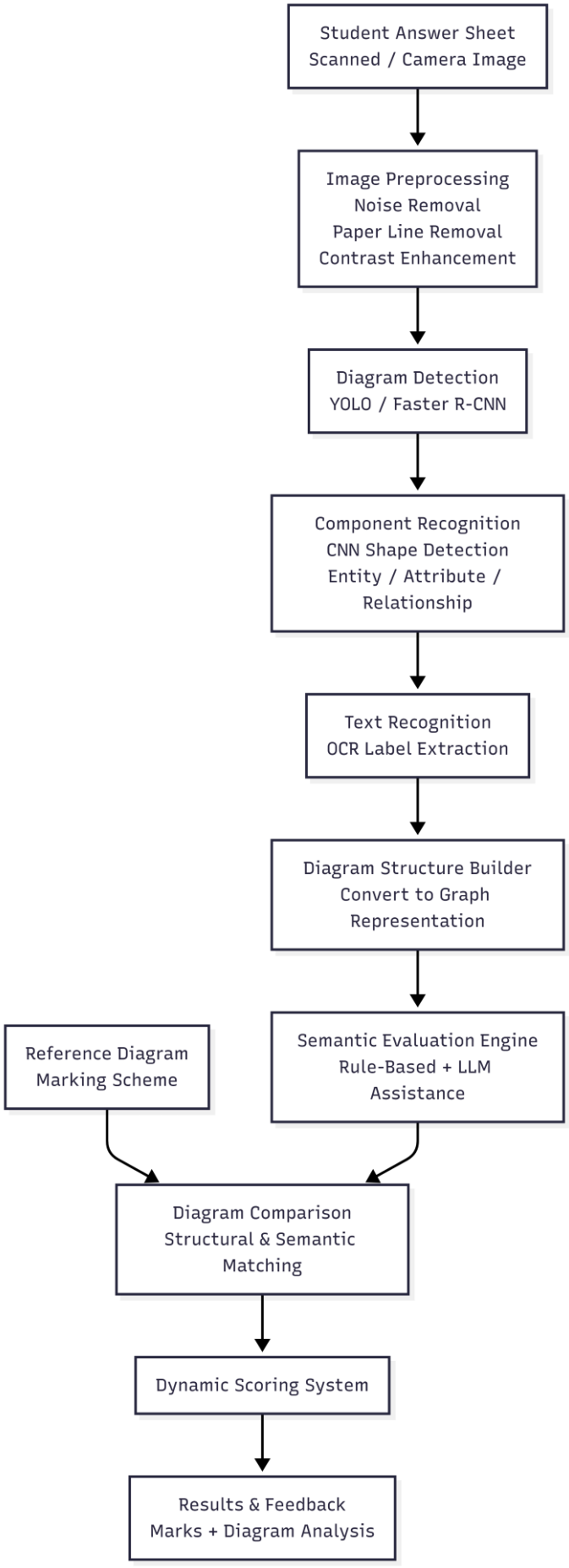
This system diagram represents an AI-powered academic evaluation platform that automates the assessment of student answers from multiple input sources.

It begins with data inputs such as physical answer sheets, historical exam papers, and student performance data. The system processes handwritten diagrams and text through stages like image preprocessing, diagram detection, OCR, and NLP-based analysis. Extracted information is then evaluated using marking rubrics, semantic comparison, and concept extraction.

A separate multimodal evaluation module analyzes video, audio, and speech inputs to assess student responses more comprehensively, including confidence scoring.

All processed data flows into an analytics engine and central database, which generate insights such as learning gaps, performance trends, and predicted outcomes. Finally, results are presented through instructor and student dashboards, enabling efficient, consistent, and data-driven academic evaluation.

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Student Answer Sheet
Scanned / Camera Image
Image Preprocessing
Noise Removal
Paper Line Removal
Contrast Enhancement
Diagram Detection
YOLO / Faster R-CNN
Component Recognition
CNN Shape Detection
Entity / Attribute /
Relationship
Text Recognition
OCR Label Extraction
Diagram Structure Builder
Convert to Graph
Representation
Reference Diagram
Marking Scheme
Semantic Evaluation Engine
Rule-Based + LLM
Assistance
Diagram Comparison
Structural & Semantic
Matching
Dynamic Scoring System
Results & Feedback
Marks + Diagram Analysis

5. USER REQUIREMENTS

Lecturers are key stakeholders because they are responsible for evaluating student exam papers and will use the system to assist in grading handwritten diagrams more efficiently and consistently. Students are also important stakeholders since their diagram answers will be analyzed by the system, and they benefit from fair, consistent, and faster evaluation of their work. Academic institutions, such as universities and colleges, are stakeholders because they manage the examination process and aim to improve the efficiency, accuracy, and scalability of assessment systems while maintaining academic standards.

5.1 Functional Requirements

- Users can submit scanned or photographed answer sheets for processing.
- Paper lines, rulers, and other unwanted marks are eliminated to enhance diagram clarity.
- Hand-drawn shapes like rectangles, circles, arrows, and decision symbols are identified.
- Labels inside diagram components are recognized using handwriting recognition.
- Relationships between diagram elements are detected to reconstruct the diagram flow.
- The extracted diagram is compared against a reference solution to check correctness.
- Alternative correct diagram structures are accepted and evaluated appropriately.
- Marks are assigned based on diagram accuracy, structure, and semantic correctness.

5.2 Non-Functional Requirements

- Security
    - The system shall ensure secure storage of student exam data.
    - Only authorized lecturers or administrators shall have access to grading results.
- Performance
    - The system shall process uploaded exam papers within an acceptable response time.
    - The system shall be capable of handling multiple exam submissions simultaneously.
- Usability
    - The system should provide a simple and intuitive interface for lecturers to upload and review exam papers.

    - The system shall display clear assessment results and diagram interpretations.
- Reliability
    - The system shall ensure consistent and accurate diagram recognition results.
    - The system should maintain stable performance during large-scale exam processing.
- Scalability
    - The system shall support increasing numbers of exam papers and users without significant performance degradation.

6. COMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the system includes universities, colleges, and educational institutions that conduct written examinations involving diagrams. The main users are lecturers and examiners who are responsible for grading large numbers of student answer sheets, as well as academic administrators who manage assessment processes. The customer persona typically includes lecturers who need a faster and more consistent way to evaluate handwritten diagrams and institutions seeking to improve the efficiency and fairness of exam assessment.

6.2 Value Proposition

The system provides an automated solution for extracting and evaluating handwritten diagrams from exam papers, reducing the time and effort required for manual grading. It improves grading consistency, accuracy, and efficiency, while allowing lecturers to focus more on teaching rather than repetitive evaluation tasks.

6.3 Revenue Model

The system can generate revenue through a software licensing or subscription model offered to educational institutions. Universities can pay an annual or semester-based subscription fee to access the system, including updates, support, and maintenance services.

6.4 Cost Estimation

The main costs include system development, AI model training, cloud infrastructure, data storage, and system maintenance. Additional costs may include server hosting, software updates, technical support, and integration with university examination systems.

6.5 Pricing Strategy

A tiered subscription pricing strategy can be used depending on the number of students or exam papers processed. For example, smaller institutions may pay a lower fee for limited usage, while larger universities may pay higher fees for full-scale deployment and advanced features.

7. BUDGET AND JUSTIFICATION

Table 3 Budget and Cost

Item	Technology / Resource	Estimated Cost (LKR)
Software and Services		
1	LLM Model (HuggingFace)	30,000.00
2	OCR Model	20,000.00
3	Cloud Storage (Azure - 10 months)	30,000.00
4	Cloud Hosting (Azure – 10 months)	55,000.00
5	Wi-Fi / broadband connection (10 months)	25,000.00
Data Collection		
6	Data Collection Materials & Documentation	5000.00
Contingency & Miscellaneous		
7	Unexpected Expenses (10% buffer)	6320.00
Total		181,500.00

8. WORK BREAKDOWN STRUCTURE (WBS)

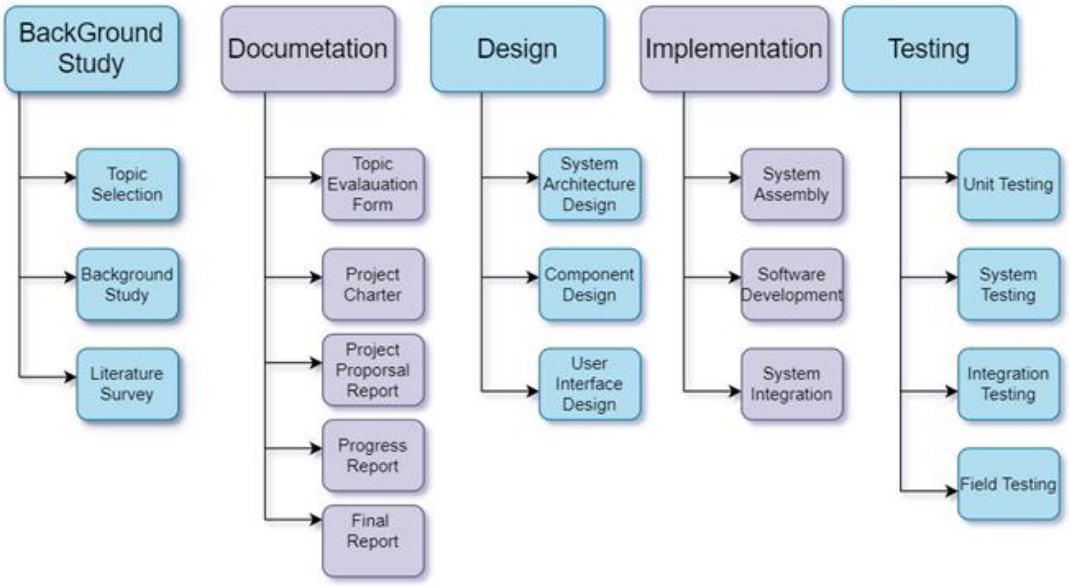


Figure 1 Work Breakdown Structure

The Work Breakdown Structure (WBS) of the project is organized into five key phases: Background Study, Documentation, Design, Implementation, and Testing. The Background Study phase includes topic selection, background research, and a literature survey. Documentation covers the creation of essential project documents such as the project charter and reports. In the Design phase, system architecture, component, and user interface designs are developed. Implementation involves assembling the system, software development, and integration. Finally, the Testing phase ensures the system's functionality and reliability through unit, system, integration, and field testing. This structured approach ensures comprehensive planning, execution, and evaluation throughout the project lifecycle.

9. GANTT CHART

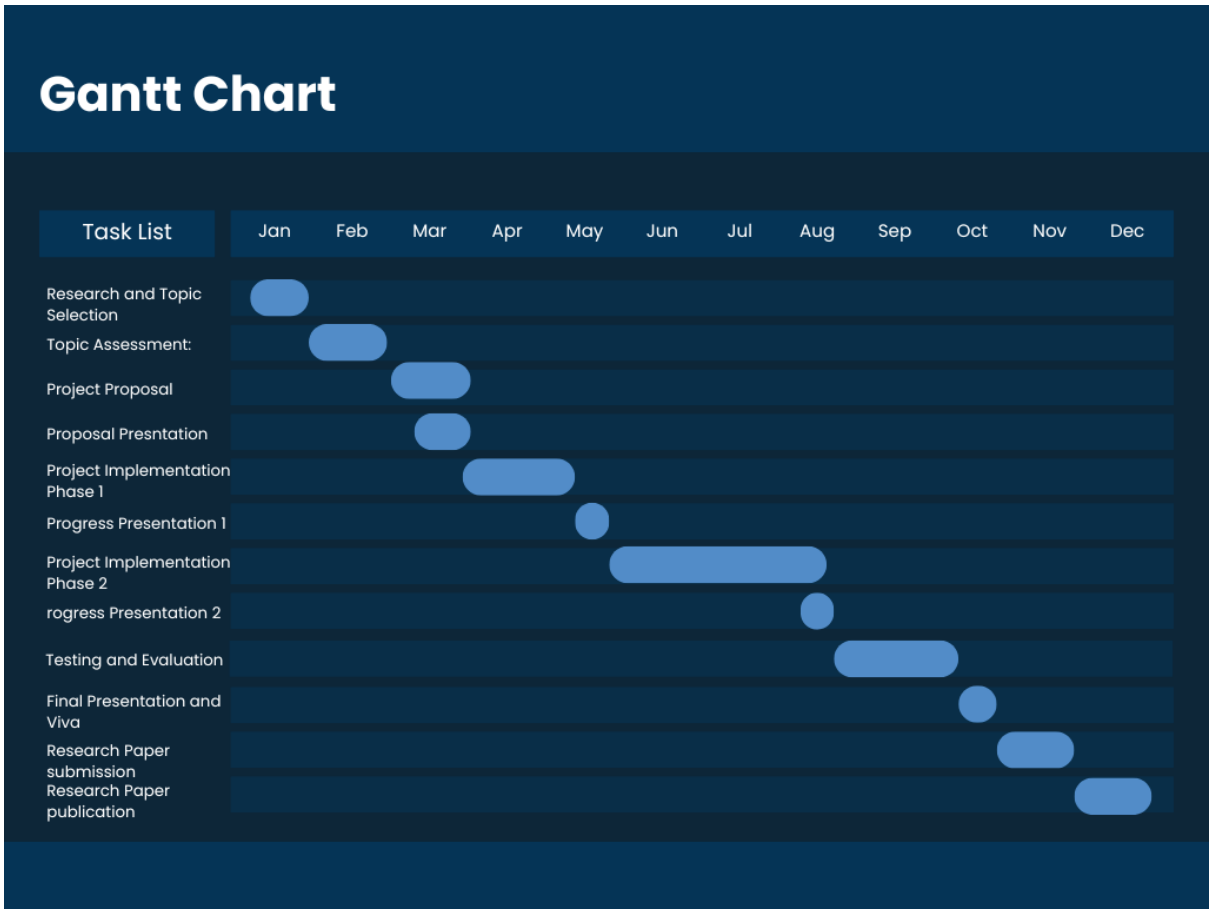


Figure 2 Gannt Chart

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11.APPENDECIES

